Sequential Monte Carlo^2 for Bayesian Filtering and Stochastic Optimal Control of a Physiological SDE

Lecture notes on the FSA-v2 model

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Generated 29 April 2026

Source: `python-smc2-filtering-control` at commit `93f93bc`

(github.com/ajaytalati/python-smc2-filtering-control)

Abstract

These notes present the mathematical theory underlying a research code base that combines particle filtering and stochastic optimal control on a three-dimensional physiological stochastic differential equation. We define the FSA-v2 model, develop the bootstrap particle filter and the SMC^2 algorithm of Chopin, Jacob and Papaspiliopoulos, recast optimal scheduling of training stimulus as posterior inference under a tempered cost likelihood, and close the loop in a model-predictive controller that interleaves a rolling SMC^2 filter with a posterior-mean control SMC^2 over an RBF schedule. We end by describing how the abstract receding-horizon loop is realised in code, by setting out a linear–quadratic–Gaussian approximation that could be added as a cheap baseline, and by flagging two outstanding analyses (Fisher information rank and Lyapunov stability) that have not yet been carried out for the model.

The audience is graduate students in applied mathematics, mathematical engineering, applied stochastics, or process engineering. Familiarity with Itô stochastic differential equations, Bayesian filtering, and basic optimal control at the level of a one-semester course is assumed; the particle-filtering and tempered-SMC machinery is built up from first principles.

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1 Introduction

A human athlete is, from the point of view of an applied mathematician, a non-linear stochastic dynamical system whose state is only ever observed through a handful of noisy proxies. Heart rate, sleep duration, perceived stress, and step counts are all loosely informative about the latent fitness, fatigue, and autonomic condition of the body, but none of them directly measures any of these quantities. The classical Banister model of training response (Banister, 1991) treats the athlete as a deterministic two-compartment system; the present notes work with a stochastic three-compartment refinement, denoted FSA-v2, in which fitness, fatigue, and an autonomic amplitude evolve as a coupled Itô stochastic differential equation driven by a single exogenous control — the daily training stimulus $\Phi(t)$.

Two questions follow naturally:

1. *Filtering.* Given a stream of multi-channel daily observations, what is the joint posterior distribution over the latent state (B, F, A) and the static parameters θ of the SDE?

2. *Control.* Given the same posterior, what schedule $\Phi : [0, T] \to [0, \Phi_{\max}]$ maximises a chosen functional of the latent trajectory, subject to a soft constraint that fatigue $F(t)$ should not exceed a physiological ceiling $F_{\max}$?

Both questions sit at the intersection of Bayesian inference and stochastic optimal control. They are usually treated as separate disciplines; one of the contributions of the underlying code base is to express both within a single sequential Monte Carlo framework. The filter is built from a bootstrap particle filter, lifted to a sampler over static parameters by the SMC^2 algorithm (Chopin, Jacob and Papaspiliopoulos, 2013) and operated in a rolling-window mode. The controller is built by recasting the cost-minimisation problem as posterior inference under

a tempered cost likelihood, in the spirit of Toussaint (2009) and the more general control-as-inference literature (Kappen, 2005; Levine, 2018). The two machines are then composed in a model-predictive controller that re-plans the next twelve hours of training stimulus from the most recent posterior.

These notes are organised so that the mathematical content can be read linearly. Section 2 defines the FSA-v2 SDE and observation model. Section 3 develops the bootstrap particle filter. The SMC^2 outer loop and its rolling-window variant are presented in Section 4. Section 5 formulates the control problem as a posterior-inference problem. Section 6 is the extended core of the document: it develops the receding-horizon principle, describes the FSA-v2 model-predictive controller in both mathematical and implementation terms, and discusses a linear–quadratic–Gaussian approximation that could be added as a cheap baseline. Section 7 reports the two headline empirical results of the code base: an open-loop gain of +28% in mean autonomic amplitude over a constant-stimulus baseline on an 84-day horizon, and a closed-loop gain of +17% in a fourteen-day rolling controller. Section 8 closes by flagging three analyses that have *not* yet been carried out on this model and that we identify as the principal open work: Fisher information rank identifiability, Lyapunov stability, and the LQG / Riccati alternative.

2 The FSA-v2 SDE model

2.1 State, drift, and diffusion

Let $(B(t),\, F(t),\, A(t))$ denote the latent state of a single individual at time $t \geq 0$, measured in days. The three components are interpreted as

- $B(t) \in [0,1]$ – a normalised *fitness* variable; the Banister chronic compartment;
- $F(t) \in [0,\infty)$ – an unnormalised *fatigue* variable; the Banister acute compartment;
- $A(t) \in [0,\infty)$ – the *amplitude* of an autonomic / circadian Stuart–Landau oscillator.

The single exogenous control input is the (non-negative) training-stimulus rate $\Phi(t) \in [0, \Phi_{\max}]$, expressed in arbitrary daily TRIMP-like units. Throughout we work in Itô form. The drift coefficients are

$$\mu(B,F) = \mu_0 + \mu_B B - \mu_F F - \mu_{FF} F^2, \tag{1}$$

$$\mathrm{d}B = \left[\, \kappa_B (1+\varepsilon_A A)\, \Phi(t) - B/\tau_B \,\right] \mathrm{d}t + \sigma_B \sqrt{B(1-B)}\, \mathrm{d}W_B, \tag{2}$$

$$\mathrm{d}F = \left[\, \kappa_F\, \Phi(t) - (1+\lambda_A A)/\tau_F \cdot F \,\right] \mathrm{d}t + \sigma_F \sqrt{F}\, \mathrm{d}W_F, \tag{3}$$

$$\mathrm{d}A = \left[\, \mu(B,F)\, A - \eta\, A^3 \,\right] \mathrm{d}t + \sigma_A \sqrt{A}\, \mathrm{d}W_A, \tag{4}$$

where W_B, W_F, W_A are independent scalar Brownian motions. The diffusion in (2) is the Jacobi diffusion on $[0,1]$ (Forman and Sørensen, 2008): the volatility vanishes at both boundaries $B=0$ and $B=1$, so the state never leaves the unit interval. The diffusions in (3) and (4) are square-root (CIR-type, after Cox, Ingersoll and Ross, 1985): the volatility vanishes at the lower boundary, so F and A remain non-negative.

The drift terms have a natural physiological reading. Equation (2) is the standard Banister chronic equation $\dot{B} = \kappa_B \Phi - B/\tau_B$, augmented by an autonomic-coupling factor $(1+\varepsilon_A A)$: when the autonomic amplitude is high the body adapts more efficiently from the same training stimulus. Equation (3) is the Banister acute equation, with the clearance rate multiplied by $(1+\lambda_A A)$: a high autonomic amplitude clears fatigue faster. Equation (4) is a Stuart–Landau cubic, in which the linear growth rate $\mu(B,F)$ depends on the slow Banister state. As μ crosses zero the deterministic skeleton undergoes a Hopf-type bifurcation: for $\mu < 0$ the origin is stable, and for $\mu > 0$ a stable limit cycle of radius $\sqrt{\mu/\eta}$ appears. The quadratic $-\mu_{FF} F^2$ term in (1) encodes *overtraining collapse*: when fatigue is excessive, the autonomic system loses its ability to oscillate.

2.2 Truth parameters

Table 1 lists the truth parameters used throughout these notes. They correspond to the canonical Banister time constants $\tau_B = 42\,\mathrm{d}$, $\tau_F = 7\,\mathrm{d}$ and to a Stuart–Landau amplitude with deterministic equilibrium $A^* \approx 1$ when $\mu \approx 0.2$.

Group	Symbol	Value	Meaning
Banister timescales	τ_B	42	chronic time constant (d)
	τ_F	7	acute time constant (d)
Banister gains	κ_B	0.012	B-gain per unit Φ
	κ_F	0.030	F-gain per unit Φ
A-coupling	ε_A	0.40	boost to B-gain at $A = 1$
	λ_A	1.00	boost to F-clearance at $A = 1$
Stuart–Landau	μ_0	0.02	baseline growth rate
	μ_B	0.30	fitness raises μ
	μ_F	0.10	linear fatigue suppression
	μ_{FF}	0.40	quadratic fatigue collapse
	η	0.20	cubic damping
Diffusion	σ_B	0.010	Jacobi noise scale
	σ_F	0.012	CIR noise scale on F
	σ_A	0.020	CIR noise scale on A

Table 1: Truth parameters of the FSA-v2 dynamics. Source: `version_2/models/fsa_high_res/_dynamics.py`.

2.3 The training-stimulus envelope

The control variable in the SDE is the instantaneous training-stimulus rate $\Phi(t)$. In practice training is delivered in discrete daily sessions concentrated in the morning, and the simulator expands a daily schedule $(\Phi_d)_{d=1,\dots,N}$ into a high-resolution envelope at fifteen-minute resolution as follows. Let $\tau_{\mathrm{burst}} = 3\,\mathrm{h}$ be a shape parameter, and let $[t_{\mathrm{wake}}, t_{\mathrm{sleep}}] = [7, 23]$ be the daily wake window (in hours). The within-day envelope is

$$e(t) \;\propto\; \begin{cases} (t - t_{\mathrm{wake}})\,\exp\!\big(-(t - t_{\mathrm{wake}})/\tau_{\mathrm{burst}}\big) & t_{\mathrm{wake}} \le t < t_{\mathrm{sleep}}, \\ 0 & \text{otherwise}, \end{cases} \tag{5}$$

which is the standard Gamma($k = 2$) shape, peaking at $t - t_{\mathrm{wake}} = \tau_{\mathrm{burst}}$. The constant of proportionality is fixed so that

$$\int_{\mathrm{day}} \Phi(t)\,\mathrm{d}t \;=\; 24\,\Phi_d, \tag{6}$$

i.e. the daily integral of the within-day envelope is preserved. The slow Banister equations (2)–(3) therefore see the same daily load whether Φ is delivered as a constant or as a morning-loaded burst, and the fast Stuart–Landau dynamics in A pick up the within-day shape.

2.4 Four-channel observation model

The state (B, F, A) is not directly observed. Instead we receive four discrete-time channels at the fifteen-minute grid:

$$\mathrm{HR}(t) \sim \mathcal{N}\big(\,\mathrm{HR}_{\mathrm{base}} - \kappa_B^{\mathrm{HR}}\, B + \alpha_A^{\mathrm{HR}}\, A + \beta_C^{\mathrm{HR}}\, C(t),\ \sigma_{\mathrm{HR}}^2\big), \qquad \text{(sleep-gated)} \tag{7}$$

$$P(\mathrm{sleep}(t) = 1 \mid A, C) = \sigma\big(\,k_C\, C(t) + k_A\, A - \tilde{c}\,\big), \qquad \text{(always observed)} \tag{8}$$

$$S(t) \sim \mathcal{N}\big(\,S_{\mathrm{base}} + k_F\, F - k_A^S\, A + \beta_C^S\, C(t),\ \sigma_S^2\big), \qquad \text{(wake-gated)} \tag{9}$$

$$\log(\mathrm{steps}(t) + 1) \sim \mathcal{N}\big(\,\mu_0^{\mathrm{step}} + \beta_B^{\mathrm{step}}\, B - \beta_F^{\mathrm{step}}\, F + \beta_A^{\mathrm{step}}\, A + \beta_C^{\mathrm{step}}\, C(t),\ \sigma_{\mathrm{step}}^2\big), \qquad \text{(wake-gated)} \tag{10}$$

where $C(t) = \cos(2\pi t + \phi)$ is a fixed deterministic circadian covariate (not a latent variable). "Sleep-gated" channels are emitted only when the sleep label of (8) is one; "wake-gated" channels only when it is zero. The collection of observation-model parameters $(\mathrm{HR}_{\mathrm{base}}, \kappa_B^{\mathrm{HR}}, \ldots, \sigma_{\mathrm{step}})$ brings the full static parameter vector θ_{dyn} for the joint state-space model to roughly 30 scalar entries. We will write $y_{1:n}$ for the multi-channel observation record up to time t_n and $p(y_{1:n} \mid \theta_{\mathrm{dyn}})$ for its marginal likelihood under the model.

2.5 Two outstanding analyses

The model defined by (1)–(4) and (7)–(10) has been validated empirically by the experiments reported in Section 7, but two analyses that we would expect to be present in a complete treatment are *not yet* part of the project. We flag them here, before any of the inference machinery is built, so that the reader does not mistake their absence for completeness.

Remark 1 (Fisher information rank identifiability — outstanding). For a parameter vector $\theta \in \mathbb{R}^d$ and observations y from the model, recall that the Fisher information matrix is

$$\mathcal{I}(\theta) \;=\; \mathbb{E}_{y \sim p(\cdot \mid \theta)}\Big[\,\nabla_\theta \log p(y \mid \theta)\, \nabla_\theta \log p(y \mid \theta)^\top\Big]. \tag{11}$$

Local identifiability of θ at a point θ^* requires that $\mathcal{I}(\theta^*)$ have full rank d; rank-deficient directions correspond to one-parameter families of models that produce indistinguishable observation distributions. For FSA-v2 with $d \approx 30$, this rank check has *not been carried out.* A future revision of these notes should evaluate $\mathcal{I}(\theta^*)$ numerically – for instance via the score function obtained by automatic differentiation of the model code – at one or more representative truth points, and either verify $\mathrm{rank}\mathcal{I}(\theta^*) = d$ or enumerate the rank-deficient directions and their physical interpretation. Practical consequences are discussed in Section 8.1.

Remark 2 (Lyapunov stability of the FSA-v2 SDE — outstanding). A stability analysis for FSA-v2 would proceed in two stages, neither of which has yet been carried out:

(a) *Deterministic skeleton.* For the zero-noise system under a constant control $\Phi \equiv \Phi_{\mathrm{const}}$, identify the equilibria $(B^*, F^*, A^*)(\Phi_{\mathrm{const}})$, compute the drift Jacobian, and check the eigenvalue signs (Lyapunov's first method). The case $A = 0$ is always an equilibrium of (4); whether it is stable depends on the sign of $\mu(B^*, F^*)$. The Stuart–Landau bifurcation in A implies that for some Φ_{const} the deterministic skeleton has an additional stable limit cycle of radius $\sqrt{\mu(B^*, F^*)/\eta}$.

(b) *Stochastic stability.* Find a Lyapunov function $V : \mathbb{R}_+^3 \to \mathbb{R}_+$ for the full SDE such that the infinitesimal generator $\mathcal{L}$ satisfies a Foster–Lyapunov drift condition $\mathcal{L}V(x) \le -\alpha V(x) + c$ on the operating domain. Such an inequality would imply exponential ergodicity and the existence of a unique invariant distribution.

Neither analysis has yet been carried out for FSA-v2; both are flagged as required next steps in Section 8.2.

3 The bootstrap particle filter

3.1 Bayesian filtering recursion

Let $X_n \in \mathbb{R}^3$ denote the latent state (B, F, A) at the nth fifteen-minute time step, and let Y_n denote the corresponding multi-channel observation. For a fixed value of the static parameter θ_{dyn}, the FSA-v2 model defines a hidden Markov model with transition density

$$X_n \mid X_{n-1} \sim p(\,\cdot \mid X_{n-1},\, \Phi_n,\, \theta_{\text{dyn}}) \tag{12}$$

implicitly through the Euler–Maruyama discretisation of (2)–(4), and observation density

$$Y_n \mid X_n \sim p(\,\cdot \mid X_n,\, C(t_n),\, \theta_{\text{dyn}}) \tag{13}$$

given by the four-channel model (7)–(10). Bayesian filtering is the recursive computation of the conditional distribution $\pi_n(\mathrm{d}x_n) := p(\mathrm{d}x_n \mid Y_{1:n},\, \theta_{\text{dyn}})$. It factorises into a one-step prediction and a Bayes update:

$$\pi_{n|n-1}(\mathrm{d}x_n) = \int p(\mathrm{d}x_n \mid x_{n-1},\, \Phi_n,\, \theta_{\text{dyn}})\, \pi_{n-1}(\mathrm{d}x_{n-1}), \tag{14}$$

$$\pi_n(\mathrm{d}x_n) \propto p(Y_n \mid x_n,\, C(t_n),\, \theta_{\text{dyn}})\; \pi_{n|n-1}(\mathrm{d}x_n). \tag{15}$$

Equations (14)–(15) have closed form only when both densities are linear-Gaussian (the Kalman case). For FSA-v2 they do not, and a Monte Carlo approximation is required.

3.2 Sequential importance resampling

The bootstrap particle filter (Gordon, Salmond and Smith, 1993; Doucet and Johansen, 2011) approximates π_n by a weighted empirical measure

$$\pi_n^N(\mathrm{d}x_n) \;=\; \sum_{i=1}^{N} W_n^{(i)}\, \delta_{X_n^{(i)}}(\mathrm{d}x_n), \qquad \sum_{i=1}^{N} W_n^{(i)} = 1, \tag{16}$$

on a population of N *state particles* $\{X_n^{(i)}\}$. The recursion has three substeps: (i) propagate each particle through the dynamics, (ii) reweight by the observation likelihood, (iii) resample to prevent weight degeneracy. This is sequential importance resampling (SIR), summarised in Algorithm 1. Resampling is performed *systematically*: with $u_0 \sim \mathcal{U}[0, 1)$, the resampled indices are obtained by walking the cumulative weight $C(j) = \sum_{i \le j} W_n^{(i)}$ at the offsets $u_k = (k + u_0)/N$ for $k = 0, \ldots, N-1$. Systematic resampling is $O(N)$, has lower Monte Carlo variance than multinomial resampling, and is deterministic given u_0.

Algorithm 1 Bootstrap SIR for the FSA-v2 filter, given θ_{dyn}

1: Sample $X_0^{(i)} \sim p(\mathrm{d}x_0 \mid \theta_{\text{dyn}})$ for $i = 1, \ldots, N$; set $W_0^{(i)} = 1/N$.
2: **for** $n = 1, 2, \ldots$ **do**
3: **Propagate.** For each i, draw $X_n^{(i)} \sim p(\,\cdot \mid X_{n-1}^{(i)},\, \Phi_n,\, \theta_{\text{dyn}})$ by one Euler–Maruyama step of (2)–(4).
4: **Reweight.** Set $\tilde{W}_n^{(i)} = W_{n-1}^{(i)}\, p(Y_n \mid X_n^{(i)},\, C(t_n),\, \theta_{\text{dyn}})$ and $W_n^{(i)} = \tilde{W}_n^{(i)} / \sum_j \tilde{W}_n^{(j)}$.
5: **ESS check.** Compute $N_{\text{eff}} = 1/\sum_i (W_n^{(i)})^2$. If $N_{\text{eff}} < N_{\text{thr}}$, resample $\{X_n^{(i)}\}$ systematically by $\{W_n^{(i)}\}$ and reset $W_n^{(i)} \leftarrow 1/N$.
6: **end for**

The implementation in `smc2fc/filtering/gk_dpf_v3_lite.py` adds two refinements that we mention but do not develop here. First, when parameter particles are coupled to the state particles in the SMC^2 outer layer of Section 4, a Liu–West shrinkage move is applied with an effective-sample-size dependent bandwidth, slowing parameter-cloud collapse. Second, when the effective sample size collapses despite resampling, an optional low-rank Sinkhorn optimal transport step blends a transport plan into the systematic resample. Both are diagnostic / numerical safeguards rather than changes to the mathematical statement of Algorithm 1.

3.3 The unbiased marginal likelihood

A key property of the bootstrap filter, which makes the SMC^2 algorithm of the next section possible, is that it produces an unbiased estimator of the marginal likelihood. With the unnormalised weights $\tilde{W}_n^{(i)}$ from line 4 of Algorithm 1, define

$$\widehat{L}_N(\theta_{\text{dyn}}) \;:=\; \prod_{n=1}^{T}\Bigl(\frac{1}{N}\sum_{i=1}^{N}\tilde{W}_n^{(i)}\Bigr). \tag{17}$$

Proposition 3 (Unbiasedness; Del Moral, 2004)**.** *For any $N \geq 1$ and any $T \geq 1$, $\mathbb{E}[\,\widehat{L}_N(\theta_{\text{dyn}})\,] = p(Y_{1:T} \mid \theta_{\text{dyn}})$.*

A proof is in Del Moral (2004, Thm. 7.4.2). For the present notes the significance of Proposition 3 is operational: it allows us to plug $\widehat{L}_N(\theta_{\text{dyn}})$ in place of the (intractable) exact likelihood $p(Y_{1:T} \mid \theta_{\text{dyn}})$ inside any sampler over θ_{dyn} and obtain a procedure that targets exactly the right posterior. This is precisely what SMC^2 does.

4 SMC^2 over parameters

4.1 The tempered posterior over θ_{dyn}

The bootstrap filter of Section 3 assumes the static parameter θ_{dyn} is known. In practice θ_{dyn} is itself the principal object of inference, with a prior $p(\theta_{\text{dyn}})$ and posterior

$$p(\theta_{\text{dyn}} \mid Y_{1:T}) \;\propto\; p(\theta_{\text{dyn}})\, p(Y_{1:T} \mid \theta_{\text{dyn}}). \tag{18}$$

A direct sampler is unavailable because $p(Y_{1:T} \mid \theta_{\text{dyn}})$ has no closed form. The SMC^2 algorithm of Chopin, Jacob and Papaspiliopoulos (2013) replaces this likelihood by the unbiased estimator $\widehat{L}_N(\theta_{\text{dyn}})$ from (17) and, for a fixed schedule $0 = \lambda_0 < \lambda_1 < \cdots < \lambda_K = 1$, defines a sequence of tempered targets

$$\pi_\lambda(\theta_{\text{dyn}}) \;\propto\; p(\theta_{\text{dyn}})^{1-\lambda}\,\widehat{L}_N(\theta_{\text{dyn}})^{\lambda}, \qquad \lambda \in [0,1]. \tag{19}$$

At $\lambda = 0$ the target is the prior; at $\lambda = 1$ it is, in expectation, the posterior. The interpolation in between bridges the two in a way that keeps the importance weights well-behaved.

4.2 Adaptive tempering and the move step

A population of M *parameter particles* $\{\theta_{\text{dyn}}^{(m)}\}_{m=1,\dots,M}$ is propagated through the schedule $\{\lambda_k\}$. At each step the unnormalised tempering weight

$$\omega_k^{(m)} \;=\; \Bigl(\frac{\widehat{L}_N(\theta_{\text{dyn}}^{(m)})}{p(\theta_{\text{dyn}}^{(m)})^{-1}}\Bigr)^{\lambda_k-\lambda_{k-1}} \tag{20}$$

is computed, and an effective sample size $M_{\text{eff}} = (\sum_m \omega_k^{(m)})^2 / \sum_m (\omega_k^{(m)})^2$ is monitored. The interval $\lambda_k - \lambda_{k-1}$ is chosen *adaptively*, by bisection, to drive M_{eff}/M to a prescribed target

(typically 0.5); this avoids both unnecessarily short bridges and weight collapse. After resampling, each parameter particle is moved by a Hamiltonian Monte Carlo (HMC) kernel that targets π_{λ_k}, with the mass matrix re-estimated from the current particle cloud. Algorithm 2 summarises the procedure.

Algorithm 2 Adaptive-tempered SMC^2 over θ_{dyn}

Sample $\theta_{\text{dyn}}^{(m)} \sim p(\theta_{\text{dyn}})$ for $m = 1, \ldots, M$; set $\lambda_0 = 0$.
For each m, run Algorithm 1 with $\theta_{\text{dyn}}^{(m)}$ to obtain $\widehat{L}_N(\theta_{\text{dyn}}^{(m)})$.
while $\lambda_k < 1$ **do**
    Solve for the smallest increment $\delta \in (0, 1 - \lambda_k]$ such that the tempering ESS ratio with weights $\omega^{(m)} \propto \widehat{L}_N(\theta_{\text{dyn}}^{(m)})^\delta$ equals the target (e.g. 0.5). Set $\lambda_{k+1} = \lambda_k + \delta$.
    Resample $\{\theta_{\text{dyn}}^{(m)}\}$ systematically by $\{\omega^{(m)}\}$.
    Move. Run a small number of HMC steps targeting $\pi_{\lambda_{k+1}}$ on each $\theta_{\text{dyn}}^{(m)}$, re-running Algorithm 1 as needed to evaluate the unbiased likelihood at proposed values.
    $k \leftarrow k + 1$.
end while

4.3 Rolling-window SMC^2 and the Schrödinger–Föllmer bridge

When data arrive sequentially, running Algorithm 2 from scratch over the full record at every new arrival is wasteful. The implementation in this code base instead operates over a sliding sequence of *windows*: with a fifteen-minute time grid, each window is one day long (96 bins), the stride between consecutive windows is twelve hours (48 bins), and the macrocycle of fourteen days yields $(14 \cdot 96 - 96)/48 + 1 = 27$ windows.

For window $w \geq 2$ the parameter cloud is *warm-started* from the posterior of window $w - 1$. The transition is mediated by a Schrödinger–Föllmer bridge between two Gaussian fits, parameterised along the Bures–Wasserstein geodesic in the manifold of positive-definite covariance matrices. Concretely, let (m_0, Σ_0) denote a Gaussian moment match to the previous window's posterior, and let (m_1, Σ_1) denote an importance-weighted moment match obtained by a single particle-filter evaluation per particle on the new window's data. For a blend parameter $\rho \in [0, 1]$ the bridge measure has mean and covariance

$$m_\rho = (1 - \rho)\, m_0 + \rho\, m_1, \tag{21}$$

$$\Sigma_\rho = \big((1 - \rho)I + \rho\, T\big)\, \Sigma_0\, \big((1 - \rho)I + \rho\, T\big)^\top, \tag{22}$$

where $T = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2}$ is the Bures–Wasserstein optimal-transport map. The default implementation uses $\rho = 0.7$, biasing the warm-start toward the new window's evidence. We do not develop the geometry here; the reader unfamiliar with it can treat the bridge as a black box that produces a sensible Gaussian initialisation for the next window's tempering schedule.

5 Control as inference

5.1 The cost functional

Sections 3 and 4 treat the dynamics parameters θ_{dyn} and the latent state X as the things to be inferred from a fixed schedule Φ. We now reverse the perspective: the dynamics parameters and the initial state are taken as given, and the schedule itself becomes the unknown.

For a fixed truth θ_{dyn}, an initial state x_0, and a candidate schedule $\Phi : [0, T] \to [0, \Phi_{\max}]$, define the trajectory cost

$$J(\Phi) \;=\; \mathbb{E}_{X|\Phi}\left[-\int_0^T A(t)\,\mathrm{d}t \;+\; \lambda_F \int_0^T \max\!\big(F(t) - F_{\max},\, 0\big)^2\,\mathrm{d}t \right], \tag{23}$$

where the expectation is over the law of $X = (B, F, A)$ under the SDE (2)–(4) driven by Φ. The first term rewards mean autonomic amplitude; the second is a soft barrier penalising trajectories that exceed the fatigue ceiling $F_{\max} = 0.40$. The barrier weight is $\lambda_F = 1$ in the reference implementation; no Φ^2 control penalty is included.

5.2 Schedule decoder

The schedule Φ is parameterised in finite dimensions by a vector $\theta_{\mathrm{ctrl}} \in \mathbb{R}^8$ of Gaussian-RBF anchor coefficients. With a fixed basis $\{\varphi_a(t)\}_{a=1}^{8}$ (eight evenly spaced Gaussians on $[0, T]$ at the discretisation grid) and a logit bias $c_\Phi := \log\!\big(\Phi_{\mathrm{def}}/(\Phi_{\max} - \Phi_{\mathrm{def}})\big)$ chosen so that $\theta_{\mathrm{ctrl}} = 0$ produces the canonical-Banister default $\Phi \equiv \Phi_{\mathrm{def}} = 1.0$, the decoder is

$$\Phi(t;\, \theta_{\mathrm{ctrl}}) \;=\; \Phi_{\max} \cdot \sigma\!\left(c_\Phi + \sum_{a=1}^{8} (\theta_{\mathrm{ctrl}})_a\, \varphi_a(t) \right). \tag{24}$$

The sigmoid envelope keeps $\Phi(t;\, \theta_{\mathrm{ctrl}}) \in [0, \Phi_{\max}]$ for every θ_{ctrl}; the search is unconstrained in $\mathbb{R}^8$.

5.3 Tempered control posterior

Following the control-as-inference tradition (Kappen, 2005; Toussaint, 2009; Levine, 2018), we define a probability measure on θ_{ctrl} by treating the negative cost as an unnormalised log-likelihood. With inverse temperature $\beta > 0$ and a Gaussian prior $\theta_{\mathrm{ctrl}} \sim \mathcal{N}(0,\, \sigma^2_{\mathrm{prior}} I_8)$, the *control posterior* is

$$q_\beta(\theta_{\mathrm{ctrl}}) \;\propto\; \exp\!\big(-\beta\, J(\theta_{\mathrm{ctrl}})\big)\, \mathcal{N}(\theta_{\mathrm{ctrl}};\, 0,\, \sigma^2_{\mathrm{prior}} I_8), \tag{25}$$

where we now write $J(\theta_{\mathrm{ctrl}}) := J(\Phi(\,\cdot\,; \theta_{\mathrm{ctrl}}))$. Sampling from q_β is performed by exactly the SMC^2 machinery of Section 4, but with the unbiased filter likelihood $\widehat{L}_N(\theta_{\mathrm{dyn}})$ replaced by the cost-likelihood $\exp(-\beta\, J(\theta_{\mathrm{ctrl}}))$. The inverse temperature β is calibrated automatically from the cost spread of the prior cloud, and the tempered schedule $\lambda \in [0, 1]$ smoothly interpolates from prior to control posterior.

5.4 Posterior mean as the controller output

Proposition 4 (Posterior mean as a regularised cost minimiser)**.** *Let q_β be the control posterior* (25) *and write $\bar{\theta} := \mathbb{E}_{q_\beta}[\theta_{\mathrm{ctrl}}]$. Then*

$$\bar{\theta} \;=\; \arg\min_{\theta \in \mathbb{R}^8} \Big\{ \mathbb{E}_{q_\beta}\big[\, \|\theta - \theta_{\mathrm{ctrl}}\|_2^2 \,\big] \Big\}, \tag{26}$$

i.e. the posterior mean is the L^2 minimiser of squared distance under the tempered measure. In particular, when J is approximately quadratic near its minimum and the prior variance is large enough that the prior contribution to $\nabla \log q_\beta$ is small, $\bar{\theta}$ sits in a neighbourhood of the unregularised minimiser $\arg\min_\theta J(\theta)$ *of size $O(\beta^{-1/2})$.*

The implementation uses $\bar{\theta}$, computed as the equally-weighted mean of the SMC^2 parameter cloud at $\lambda = 1$, as the controller output. The corresponding schedule is then $\Phi(\,\cdot\,; \bar{\theta})$ via (24). We will see in Section 7 that this point output is sufficient in practice, despite throwing away the higher-order moments of q_β; Section 6.7 discusses why.

6 Receding-horizon model-predictive control

This section is the conceptual core of the document. It develops the finite-horizon stochastic optimal control problem, the receding-horizon principle, the various stochastic and output-feedback variants, and then specialises everything to the FSA-v2 closed-loop model-predictive controller in both mathematical and implementation form. A closing subsection sketches a linear–quadratic–Gaussian approximation that could be added as a cheap baseline.

6.1 Finite-horizon stochastic optimal control

Consider a discrete-time stochastic system on a state $X_k \in \mathcal{X}$ and a control $U_k \in \mathcal{U}$, with transition kernel $X_{k+1} \sim p(\cdot \mid X_k, U_k)$, a stage cost $\ell : \mathcal{X} \times \mathcal{U} \to \mathbb{R}$, and a terminal cost $V_f : \mathcal{X} \to \mathbb{R}$. The *open-loop* finite-horizon problem of length H is

$$\min_{u_{0:H-1} \in \mathcal{U}^H} \mathbb{E}\left[\sum_{k=0}^{H-1} \ell(X_k, u_k) + V_f(X_H) \,\middle|\, X_0 = x \right], \tag{27}$$

with the expectation taken over the joint law of $X_{1:H}$ given the chosen sequence $u_{0:H-1}$. Denote its optimal value by $V_H(x)$ and a minimising sequence by $u^*_{0:H-1}(x)$. The mapping $x \mapsto u^*_0(x)$ is the open-loop optimal first-action policy.

For FSA-v2 the state $X_k = (B_k, F_k, A_k)$ lives in $\mathcal{X} = [0,1] \times [0,\infty)^2$, and the "control sequence" is the eight-dimensional RBF-coefficient vector θ_{ctrl} that parameterises the entire schedule $\Phi(\,\cdot\,; \theta_{\text{ctrl}})$ over $[0,T]$ via (24). The discrete-time abstract notation (27) therefore maps onto the continuous schedule of Section 5 by a one-to-one identification between admissible $u_{0:H-1}$ and admissible θ_{ctrl}. The expectation in (27) is the same one already written out in (23).

6.2 The receding-horizon principle

In *receding-horizon* or *model-predictive* control, the open-loop problem (27) is solved repeatedly on a sliding window. At each replan time t_n, given a current state estimate $\widehat{x}_n$, one

1. solves (27) for $u^*_{0:H-1}(\widehat{x}_n)$;
2. applies only the first action $u^*_0(\widehat{x}_n)$ over a stride of length $\Delta < H$;
3. observes new data, updates the state estimate to $\widehat{x}_{n+1}$, and returns to step 1.

The resulting closed-loop policy is a function of $\widehat{x}_n$ alone, so the controller is implementable. Stability of the closed loop is delicate. Mayne and Rawlings (2000) show that under suitable assumptions on the horizon length and the terminal cost / terminal set, the closed-loop state remains in a positively invariant set and converges to a neighbourhood of a desired equilibrium. For FSA-v2 we rely on the *long-horizon* variant of that theory: the planning horizon H is chosen long enough relative to the slowest time constant $\tau_B = 42$ days that the truncation error from neglecting trajectory beyond H is small.

6.3 Stochastic and output-feedback variants

Three flavours of stochastic MPC are commonly distinguished.

(i) *Certainty-equivalent* MPC. Replace the expectation in (27) by evaluation at a point estimate $\hat{X} = \mathbb{E}[X \mid \widehat{x}_n]$, and solve the resulting deterministic problem. Cheap, but discards uncertainty information.

(ii) *Expected-cost* (stochastic) MPC. Keep the expectation, and approximate it by Monte Carlo (sampling trajectories of the stochastic dynamics) or by scenario trees.

(iii) *Output-feedback* MPC. When X_k is not observed directly, the controller acts on a state estimate $\widehat{x}_n$ produced by an external estimator (Kalman, particle filter, SMC^2, etc.). For nonlinear non-Gaussian systems no *separation principle* holds in general — the joint design of the estimator and the controller is part of the modelling, not a free composition.

The FSA-v2 pipeline is in class (ii) and (iii) simultaneously. The state is unobserved (only the four-channel proxies of (7)–(10) are available), so a state estimator is required: this is the rolling-window SMC^2 filter of Section 4. The expectation in the cost (23) is kept, evaluated by Monte Carlo on the SDE under common random numbers, and minimised in θ_{ctrl}-space by the control SMC^2 of Section 5. The full closed-loop policy is therefore the composition

$$\text{(observations)} \xrightarrow{\text{rolling SMC}^2\text{ filter}} \widehat{x}_n \xrightarrow{\text{control SMC}^2\text{, posterior mean}} \bar{\theta} \xrightarrow{(24)} \Phi(\,\cdot\,;\bar{\theta}) \xrightarrow{\text{plant}} \text{(new observations)}.$$

There is no separation principle here. The fact that this composition works at all is an empirical claim, validated in Section 7.

6.4 The FSA-v2 MPC pipeline — mathematical statement

We now write down the closed-loop algorithm. Let $\Delta_w = 96$ bins be the filter window length (one day), $\Delta_s = 48$ bins the stride (twelve hours), and $H_T = 14 \cdot 96 = 1344$ bins the macrocycle (fourteen days), so that the number of windows is $N_w = (H_T - \Delta_w)/\Delta_s + 1 = 27$.

Algorithm 3 FSA-v2 closed-loop MPC

1: Initialise the plant at (B_0, F_0, A_0); initialise an empty observation buffer $Y_{1:0} = \emptyset$.
2: **for** $n = 1, \ldots, N_w$ **do**
3: **Filter update.** If $n = 1$, run cold-start SMC^2 (Algorithm 2) on $Y_{(n-1)\Delta_s+1:(n-1)\Delta_s+\Delta_w}$ to produce a posterior over θ_{dyn} for the first window; otherwise warm-start from the previous window's posterior via the Schrödinger–Föllmer bridge of Section 4.
4: **State extraction.** For a small set of θ_{dyn} particles drawn from the posterior, run the bootstrap filter (Algorithm 1) forward to the stride endpoint and compute the importance-weighted mean of the resulting (B, F, A); call this $\widehat{x}_n$.
5: **Control plan.** Run the control SMC^2 of Section 5, conditioned on $\widehat{x}_n$ and the posterior-mean dynamics parameters $\bar{\theta}_{\text{dyn}}$, to obtain the posterior-mean schedule coefficient $\bar{\theta}_{\text{ctrl}}$.
6: **Apply.** Decode $\Phi(\,\cdot\,;\bar{\theta}_{\text{ctrl}})$ via (24) and apply only its first stride (length Δ_s) to the plant; collect the resulting observations and append them to the buffer.
7: **end for**

The horizon over which the controller plans (line 5) is the full macrocycle remainder $H_T - (n-1)\Delta_s$, but only the first Δ_s bins of the plan are committed (line 6). This is the receding-horizon truncation of Section 6.2 applied to the FSA-v2 problem. Replanning occurs every twelve hours, at the wake-window boundary so that the morning-loaded burst envelope of (5) is delivered in full each day.

6.5 Code-level structure of the MPC implementation

This subsection departs from the usual mathematical idiom and describes how the abstract loop of Section 6.4 is realised in the source code of the project. The reader who only wants the mathematics may skip to Section 6.6; the reader who intends to run, modify, or extend the controller will find this useful. We name three artefacts.

(a) The plant interface, `StepwisePlant`. The "plant" in stochastic optimal control is whatever object the controller acts upon to produce new observations. In the code base it is

the class `StepwisePlant` in `version_2/models/fsa_high_res/_plant.py`. A `StepwisePlant` carries the hidden true state (B, F, A), the true dynamics parameters, and a global bin counter; its single mutating method is

> `advance(stride_bins, Phi_daily)` $\rightarrow$ `(trajectory, obs_HR, obs_sleep, obs_stress, obs_steps)`.

A call to `advance` expands the supplied daily-Φ array into the morning-loaded sub-daily envelope of (5) (preserving $\int_{\text{day}} \Phi \, \mathrm{d}t = 24\,\Phi_d$ exactly), runs Euler–Maruyama for `stride_bins` fifteen-minute steps, samples the four-channel observations of (7)–(10), and updates the internal state. Pedagogically, the `StepwisePlant` is the digital embodiment of the "real world" against which the controller is evaluated; mathematically, a single call samples one transition of the joint state-observation process under the chosen control over a stride of length Δ_s.

(b) The MPC driver, `bench_smc_full_mpc_fsa.py`. The receding-horizon orchestrator lives in `version_2/tools/bench_smc_full_mpc_fsa.py`. Its execution sequence implements Algorithm 3 step by step. For each window n it

1. runs the rolling-window filter by calling `run_smc_window` (cold-start, $n = 1$) or `run_smc_window_bridge` with a Schrödinger–Föllmer base measure (warm-start, $n \geq 2$);
2. extracts the posterior-mean dynamics parameters and the smoothed latent state $\widehat{x}_n$ at the end of the window, the latter computed by running Algorithm 1 forward on a small set of θ_{dyn} particles drawn from the posterior and taking the weighted mean of the resulting (B, F, A) samples;
3. assembles a `ControlSpec` (cost functional, horizon length, current state estimate, RBF basis);
4. calls the control SMC^2 loop in `smc2fc/control/tempered_smc_loop.py`, which returns a posterior-mean schedule coefficient $\bar{\theta}_{\text{ctrl}}$ and a corresponding Φ schedule;
5. calls `plant.advance(stride_bins, Phi_daily)` to apply only the first stride of that schedule;
6. appends the resulting observations to the data buffer and shifts the window.

(c) The posterior-mean handoff between windows. The smoothed state $\widehat{x}_n$ extracted in step 2 above is what initialises the next window's particle filter. Consecutive windows are therefore connected *at the state level* by a single point estimate (the importance-weighted smoothed mean) and *at the parameter level* by the SF Bures–Wasserstein bridge of Section 4. This is the practical realisation of the warm-start that gives the rolling SMC^2 its computational tractability: at fourteen days of fifteen-minute data the cold-start cost would be 27× larger.

Table 2 collects the mapping between the steps of Algorithm 3 and the source files that realise them.

6.6 Recursive feasibility and the F-barrier

The fatigue ceiling $F_{\max} = 0.40$ is enforced through the soft-barrier penalty $\lambda_F \cdot \max(F - F_{\max}, 0)^2$ in the cost (23), not through a hard constraint. The choice is deliberate. Under stochastic dynamics with state-dependent diffusion and unbounded driving noise, an exact hard constraint $F(t) \leq F_{\max}$ cannot be guaranteed by any controller without making the admissible-control set empty for some achievable initial conditions; this is the standard *recursive feasibility* difficulty in stochastic MPC. A soft penalty preserves recursive feasibility – there is always a feasible

Step in Algorithm 3	Concept	Source
3	rolling SMC^2 filter	`smc2fc/core/tempered_smc.py`
3	SF Bures–Wasserstein bridge	`smc2fc/core/sf_bridge.py`
3	inner bootstrap filter	`smc2fc/filtering/gk_dpf_v3_lite.py`
4	smoothed-state extraction	`version_2/tools/bench_smc_full_mpc_fsa.py`
5	control SMC^2 outer loop	`smc2fc/control/tempered_smc_loop.py`
5	cost functional, RBF decoder	`version_1/models/fsa_high_res/control.py`
6	plant advance	`version_2/models/fsa_high_res/_plant.py`
2,6	sub-daily Φ envelope	`version_2/models/fsa_high_res/_phi_burst.py`

Table 2: Mapping from Algorithm 3 to the source code.

schedule – and produces the constraint approximately, with the violation rate decreasing as λ_F is increased. Empirically (see Section 7) the closed-loop F-violation rate is 1.28% on the 84-day open-loop benchmark and 0.00% on the 14-day closed-loop benchmark, well within the 5% acceptance gate.

6.7 Robustness and the flat-cost-surface argument

The controller of Section 6.4 commits the posterior mean $\bar{\theta}_{\text{ctrl}}$ rather than a sample from the control posterior or the full distribution. One might worry that this discards useful information, especially when the filter posterior over θ_{dyn} is uncertain or biased. Empirically the policy is nevertheless robust to filter parameter drift, and the explanation is geometric: near the optimum, the cost $J(\theta_{\text{ctrl}})$ is locally flat in the directions explored by the SMC^2 cloud. A first-order Taylor expansion of J at the posterior mean gives, for a perturbed estimate $\bar{\theta} + \delta$,

$$J(\bar{\theta} + \delta) \;\approx\; J(\bar{\theta}) + \tfrac{1}{2}\delta^\top H \delta, \qquad H = \nabla^2 J(\bar{\theta}), \tag{28}$$

so the additional cost from using $\bar{\theta}$ in place of an unknown "oracle" optimum scales with the Hessian. When H is small in the directions of greatest filter uncertainty, the closed-loop performance is correspondingly insensitive to filter accuracy. Section 7 reports closed-loop performance on a record where only 3 out of 26 estimable parameters pass a strict coverage gate; the controller nevertheless achieves a +17% amplitude gain. The flat-cost-surface argument is the mechanism.

6.8 An exact-approximate alternative: LQG and the differential Riccati equation

We close Section 6 by sketching a second controller that could be added to the project as a cheap baseline or as a warm-start for the SMC^2 control of Section 5. The sketch is deliberately complete enough to make the assumptions and limitations explicit; it is not yet implemented in the code base, and is flagged as required next work in Section 8.3.

Linearisation around an operating point. Three assumptions reduce the FSA-v2 problem to one with a closed-form optimal feedback law.

Assumption 5. There is a nominal trajectory $(x^*(t), \Phi^*(t))$ on $[0, T]$, taken in the simplest case to be the deterministic-skeleton equilibrium $(B^*, F^*, A^*)(\Phi_{\text{const}})$ under a constant control. With $\tilde{x} = x - x^*$ and $\tilde{\Phi} = \Phi - \Phi^*$, define the linearised drift coefficients

$$A_{\text{lin}}(t) = \left.\frac{\partial f}{\partial x}\right|_*, \qquad B_{\text{lin}}(t) = \left.\frac{\partial f}{\partial \Phi}\right|_*, \tag{29}$$

where f is the drift of (2)–(4).

The cubic Stuart–Landau term $-\eta A^3$ is linearised away under Assumption 5, so the bistable / limit-cycle behaviour of the autonomic amplitude is lost. The linearisation is valid only in a neighbourhood of the operating point.

Assumption 6. The state-dependent Jacobi and CIR diffusions in (2)–(4) are replaced by constant-coefficient Gaussian noise $\bar{\sigma}\,\mathrm{d}W$ evaluated at the operating point.

The $[0,1]$ and $[0,\infty)$ state constraints are no longer enforced implicitly by the noise vanishing at the boundary, so trajectories may leave the physiological domain. Acceptable when fluctuations around the operating point are small relative to the boundary distances.

Assumption 7. The cost (23) is replaced by the quadratic surrogate

$$J_{\mathrm{LQ}}(\tilde{\Phi}) \;=\; \mathbb{E}\left[\int_0^T (\tilde{x}^\top Q\tilde{x} + \tilde{\Phi}^\top R\tilde{\Phi})\,\mathrm{d}t + \tilde{x}(T)^\top Q_T\tilde{x}(T)\right], \tag{30}$$

with $Q \succeq 0$, $R \succ 0$, $Q_T \succeq 0$.

To approximate the $-\int A\,\mathrm{d}t$ reward, one chooses $x_{\mathrm{ref}} \neq x^*$ shifted toward higher A and penalises $\tilde{x} = x - x_{\mathrm{ref}}$ with Q giving most weight to the A component; this preserves $Q \succeq 0$. The soft fatigue barrier $\lambda_F \cdot \max(F - F_{\max}, 0)^2$ is replaced by a quadratic in $F - F^*$, which penalises both directions symmetrically – too restrictive on the rest side and too lax on the over-fatigue side near the barrier.

The differential Riccati equation. Under Assumptions 5–7, the optimal feedback law for the linearised system is the time-varying linear policy

$$\tilde{\Phi}^*(t) \;=\; -K(t)\,\tilde{x}(t), \qquad K(t) \;=\; R^{-1}\,B_{\mathrm{lin}}(t)^\top\,P(t), \tag{31}$$

where $P : [0,T] \to \mathbb{R}^{3\times 3}$, $P(t) \succeq 0$, solves the *differential Riccati equation*

$$-\dot{P}(t) \;=\; A_{\mathrm{lin}}(t)^\top P(t) + P(t)\,A_{\mathrm{lin}}(t) - P(t)\,B_{\mathrm{lin}}(t)\,R^{-1}\,B_{\mathrm{lin}}(t)^\top P(t) + Q, \qquad P(T) = Q_T. \tag{32}$$

For a time-invariant linearisation (constant A_{lin}, B_{lin}) and an infinite horizon, P solves the *algebraic* Riccati equation $0 = A_{\mathrm{lin}}^\top P + P\,A_{\mathrm{lin}} - P\,B_{\mathrm{lin}}\,R^{-1}\,B_{\mathrm{lin}}^\top P + Q$. Under Assumptions 5–6 the state-estimation problem also linearises and is solved by the Kalman–Bucy filter; the LQG controller is then the certainty-equivalent composition of Kalman–Bucy filter and LQR feedback, and the (linear-Gaussian) separation principle holds.

Where this would fit in the project. A complete LQG companion to the SMC^2 controller would consist of a new module `smc2fc/control/lqg/` containing

1. automatic linearisation of the drift in `version_2/models/fsa_high_res/_dynamics.py` around a user-supplied operating point, via JAX automatic differentiation;
2. a numerical solver for (32) by backward integration on the same fifteen-minute time grid used elsewhere in the project;
3. a `LQGController` class emitting the feedback gain $K(t)$ of (31) and a corresponding Φ schedule on the project's time grid.

The natural uses are (i) as a *cheap baseline* against which the SMC^2 controller of Section 5 is benchmarked — does the non-linear posterior-mean controller actually outperform the linear LQG approximation, and by how much? — and (ii) as a *warm-start* for the control posterior, by least-squares projection of the LQG schedule onto the RBF basis to give an initial θ_{ctrl} around which the SMC^2 then explores. None of this currently exists in the repository; we return to it in Section 8.3.

7 Results

The two headline empirical results of the project are an open-loop benchmark on the canonical 84-day horizon (one chronic time constant of 42 days, doubled) and the closed-loop fourteen-day rolling controller. Both compare the SMC^2 schedule to the canonical constant baseline $\Phi \equiv 1.0$. Headline numbers are summarised in Table 3.

Benchmark	Horizon	$\bar{A}$ (SMC^2)	$\bar{A}$ (baseline)	gain	F-violation
Open-loop, Stage D	84 d	0.645	0.503	+28%	1.28%
Closed-loop MPC, Stage E5	14 d	0.0964	0.0823	+17%	0.00%

Table 3: Headline empirical results. $\bar{A}$ denotes the mean autonomic amplitude $\frac{1}{T}\int_0^T A(t)\,\mathrm{d}t$ averaged over Monte Carlo trajectories. Sources: `version_1/outputs/fsa_high_res/RESULT.md` (Stage D), `version_2/outputs/fsa_high_res/RESULT.md` (Stage E5).

7.1 Stage D: open-loop schedule on the 84-day horizon

In the open-loop benchmark, the dynamics parameters θ_{dyn} are treated as known at their truth values, the initial state is fixed at $(B_0, F_0, A_0) = (0.05, 0.30, 0.10)$ (a de-trained athlete with high residual fatigue), and the SMC^2 control of Section 5 is run *once* over the full 84-day horizon. The output is a single posterior-mean schedule $\Phi(\,\cdot\,;\bar{\theta})$.

The optimised schedule has mean $\Phi \approx 1.96$ over the horizon and a front-loaded periodisation profile (heavy training for the first ~ 60 days, taper thereafter). Mean amplitude $\bar{A} = 0.645$ exceeds the constant baseline $\bar{A} = 0.503$ by 28%, exceeds the sedentary reference $\bar{A} = 0.209$ by a factor of 3.08, and the soft fatigue barrier is violated only 1.28% of the time — well within the 5% acceptance gate of Section 6.6. All four acceptance gates (mean-A matches baseline within 3%, mean-A $\geq 1.40\times$ sedentary, mean-$\Phi \in [0.5, 2.5]$, F-violation $\leq 5\%$) pass.

7.2 Stage E5: closed-loop MPC on the 14-day macrocycle

In the closed-loop benchmark the dynamics parameters are no longer known to the controller. Algorithm 3 is run on a 14-day record of four-channel observations, with a one-day filter window, twelve-hour stride, and 27 replans. The plant is the `StepwisePlant` of Section 6.5.

Mean amplitude under the closed-loop policy is $\bar{A} = 0.0964$, against $\bar{A} = 0.0823$ for the constant baseline $\Phi \equiv 1.0$ on the same 14-day horizon. (The shorter horizon explains the smaller absolute numbers compared to Stage D: the integral $\int_0^T A(t)\,\mathrm{d}t$ has not yet converged to the asymptotic mean.) The +17% gain is achieved with a remarkable schedule shape: the controller commits a *rest-leaning* Φ in the range 0.25–0.31 across all 27 strides, against the canonical baseline of 1.0. Physiologically, this is the controller exploiting the fact that the simulated athlete starts with high residual fatigue ($F_0 = 0.30$): the optimal short-horizon strategy is to clear fatigue rather than to add training stimulus, which unlocks the Stuart–Landau growth-rate $\mu(B, F)$ and lets A rise from 0.10 to ~ 0.20 over the macrocycle.

The fatigue ceiling is never violated (0.00%). Wall-clock cost on a single GPU is 41 minutes for the full closed loop. The closed-loop policy is robust despite imperfect filter coverage — on this record only 3 of 26 estimable parameters meet a strict $\geq 5/6$ coverage gate at the window level — consistent with the flat-cost-surface argument of Section 6.7.

A trajectory plot of $A(t)$ under both policies, and a plot of the SMC^2 posterior-mean schedule, are included as Figure placeholders here; they can be regenerated from the artefacts in `version_2/outputs/fsa_high_res/` by re-running `tools/bench_smc_full_mpc_fsa.py`.

Figure placeholder: $A(t)$ trajectory under closed-loop MPC (blue) and constant baseline $\Phi \equiv 1$ (grey), with the discovered rest-leaning Φ schedule overlaid.

8 Discussion and open questions

The empirical results of Section 7 show that the SMC^2 filter and controller, composed in receding-horizon form, produce a closed-loop policy that improves mean autonomic amplitude by +17% over a sensible constant baseline on the FSA-v2 model, with no fatigue-ceiling violations. This is encouraging, but the present treatment is incomplete in three specific and important ways. Each of the three subsections below identifies one outstanding analysis that should be carried out before the model is used in a setting where its properties matter — for instance, in a real-time clinical setting where mis-identification of a parameter could mean prescribing the wrong training load.

8.1 Outstanding analysis: Fisher information rank identifiability

Remark 1 of Section 2.5 introduced the Fisher information matrix (11) and pointed out that the rank check on $\mathcal{I}(\theta^*)$ has not been carried out for FSA-v2. We now sketch what such an analysis would look like and what its consequences might be.

Procedure. Pick a representative truth point θ^* (the parameter vector of Table 1 extended with the observation-model coefficients). Generate L independent simulated trajectories of the model at θ^*, each producing a record $y_{1:T}^{(\ell)}$ of four-channel observations on the project's fifteen-minute time grid. For each trajectory, evaluate the score $s^{(\ell)}(\theta) = \nabla_\theta \log p(y_{1:T}^{(\ell)} \mid \theta)$ at $\theta = \theta^*$ by automatic differentiation through the generative code (the JAX dynamics solver of `version_2/models/fsa_high_res/_dynamics.py` together with the observation model in `simulation.py`). Approximate the Fisher information by the empirical second moment $\widehat{\mathcal{I}}(\theta^*) = L^{-1} \sum_\ell s^{(\ell)}(\theta^*)\, s^{(\ell)}(\theta^*)^\top$ and read off its singular-value decomposition. Repeat for several truth points to check that the conclusions are not parameter-specific.

Possible findings. If $\operatorname{rank}\widehat{\mathcal{I}}(\theta^*) = d$ and the smallest singular value is bounded away from zero, the model is locally identifiable at θ^* and inference is well-posed. If instead the rank is deficient by r, the right-singular vectors corresponding to the smallest r singular values give the directions in parameter space along which the likelihood is flat. Each such direction should then be examined physically: it might indicate (i) a collinearity between two physiological parameters that no four-channel observation set could resolve, in which case one of them must be fixed or tightly priored; (ii) a redundancy in the observation model itself (for instance, that HR and stress carry overlapping information about B and F); or (iii) a genuine modelling ambiguity that should be removed by reparameterisation. Consequences for the SMC^2 filter include the possibility of multimodal posteriors, slower mixing of the HMC moves of Section 4, and degraded robustness of the closed-loop policy in the directions that the filter cannot constrain.

8.2 Outstanding analysis: Lyapunov stability

Remark 2 flagged the Lyapunov analysis as an outstanding piece of work in two parts: stability of the deterministic skeleton, and stochastic stability of the full SDE.

Deterministic skeleton. For a constant control $\Phi = \Phi_{\text{const}}$, the steady-state $A^* = 0$ branch satisfies $B^* = \kappa_B(1 + \varepsilon_A \cdot 0)\Phi_{\text{const}}\, \tau_B = \kappa_B \tau_B \Phi_{\text{const}}$ and $F^* = \kappa_F\, \tau_F\, \Phi_{\text{const}}$. At this branch the Stuart–Landau growth rate is $\mu(B^*, F^*) = \mu_0 + \mu_B B^* - \mu_F F^* - \mu_{FF}(F^*)^2$, and the deterministic A-equation is locally $\dot{A} = \mu(B^*, F^*)\, A$. The $A = 0$ branch is therefore stable when

$\mu(B^*, F^*) < 0$ and unstable when $\mu(B^*, F^*) > 0$. In the latter regime an additional stable limit cycle of radius $\sqrt{\mu(B^*, F^*)/\eta}$ appears. The complete analysis would chart the bifurcation locus $\{\Phi_{\text{const}} : \mu(B^*(\Phi_{\text{const}}), F^*(\Phi_{\text{const}})) = 0\}$ and identify the critical training stimulus at which the autonomic amplitude switches from quiescent to oscillatory.

Stochastic stability. The natural candidate Lyapunov function on $\mathbb{R}^3_+$ is the quadratic $V(B, F, A) = aB^2 + bF^2 + cA^2$, with $a, b, c > 0$ to be chosen. Under Itô calculus,

$$\begin{aligned}\mathcal{L}V(B, F, A) = {} & 2aB\big[\kappa_B(1 + \varepsilon_A A)\Phi - B/\tau_B\big] + a\,\sigma_B^2\,B(1 - B) \\ & + 2bF\big[\kappa_F\Phi - (1 + \lambda_A A)/\tau_F \cdot F\big] + b\,\sigma_F^2\,F \\ & + 2cA\big[\mu(B, F)\,A - \eta A^3\big] + c\,\sigma_A^2\,A.\end{aligned}$$

The terms $-2aB^2/\tau_B$, $-2bF^2(1 + \lambda_A A)/\tau_F$ and $-2c\eta A^4$ are dissipative; the remaining terms grow at most linearly or quadratically in the state. For sufficiently large a, b, c and appropriate operating range one would hope to establish a Foster–Lyapunov drift inequality $\mathcal{L}V \le -\alpha V + c'$ on $\{V \le R\}$ for some R, which would imply exponential ergodicity and the existence of a unique invariant distribution. The argument is delicate near the Stuart–Landau bifurcation, where the A-direction is weakly stabilising or weakly unstabilising; a piecewise Lyapunov function that switches between regimes may be required. None of this has yet been carried out.

8.3 Outstanding analysis: LQG / Riccati exact-approximate alternative

Section 6.8 sketched a linear–quadratic–Gaussian controller for FSA-v2 under Assumptions 5–7. None of the corresponding code is currently in the repository. We identify three concrete pieces of work that together would deliver the LQG companion to the SMC^2 controller:

1. Implement automatic linearisation of the drift around a user-supplied operating point, using the existing JAX dynamics in `version_2/models/fsa_high_res/_dynamics.py`; expose the time-varying $A_{\text{lin}}(t), B_{\text{lin}}(t)$ of (29) on the same fifteen-minute time grid used by the rest of the project.

2. Implement a backward-integration solver for the differential Riccati equation (32) on the same time grid.

3. Build an `LQGController` class that emits the feedback gain $K(t)$ of (31) and a corresponding Φ schedule on the time grid.

Two scientific questions then become accessible. *Does the LQG schedule, evaluated as an open-loop control on the FSA-v2 model, achieve mean amplitude comparable to the SMC^2 schedule on the 84-day benchmark of Section 7?* If yes, the non-linear SMC^2 machinery may be unnecessary in steady-state regimes; if no, the gap quantifies the value of capturing the cubic Stuart–Landau non-linearity. *Does using the LQG schedule as a warm-start for the SMC^2 control of Section 5 reduce wall-clock time to solution?* A reasonable hypothesis is that LQG provides an excellent initial θ_{ctrl} in the linear regime and only marginal benefit where the non-linearity matters; experimentally testing this is the natural next step.

8.4 Other open questions

A few smaller points deserve mention but are not, in the authors' view, on the same level of importance as the three analyses above.

- *Horizon length.* The Stage E5 result uses a fourteen-day macrocycle. The Stage D open-loop result uses eighty-four days. It is not yet known how the closed-loop gain scales with horizon length; a horizon sweep is straightforward to set up.

- *Posterior shape vs. posterior mean.* The controller of Section 6.4 uses only the posterior mean $\bar{\theta}_{\text{ctrl}}$. The flat-cost-surface argument of Section 6.7 explains why this is sometimes sufficient, but in regimes where the cost surface is curved the full posterior carries useful information. A risk-sensitive variant that minimises a quantile of the cost rather than the mean would be the natural extension.
- *Parameter drift across windows.* The Bures–Wasserstein bridge of Section 4 produces a sensible Gaussian-fit warm-start, but does not protect against *slow* parameter drift in the truth (e.g. the athlete's time constants change with training age). A more principled approach would augment the static parameters with a slow random-walk component and let the filter track the drift explicitly.

The empirical successes reported in Section 7 should be read alongside the three outstanding analyses above. The model is empirically useful; it is not yet mathematically complete.